Physical Computing

Syllabus & Overview

@tamberg & @dusjagr

* Min res 56x16 char

Contents

Description

- Module phycom

* https://www.fhnw.ch/de/studium/module/9880131

Language

- Inputs in class are in German (DE)
- Course materials are in English (EN)
- Your contribution can be in DE or EN

Credits

- 3 ECTS, 2 * 5 days, 8 + 1 hours a day
- Presence, active participation
- DIY-project, teams of 2
- Pass/fail, no grades

Date

- Week 1, KW24, 08.06. – 12.06.2026
- Week 2, KW25, 15.06. – 19.06.2026

Time

- 09:15 – 12:00, 3 hours in the morning
- 13:15 – 18:00, 5 hours in the afternoon
- 15 minute breaks, roughly once an hour

Location

- Input, usually at the classroom, Room 4.513
- Making, usually at the makerspace, Room 4.125

Presence

> Students are expected to be present for the entire two weeks

- The course consists of inputs and hands-on making sessions
- We will use specific tools available in a typical makerspace
- We count on you all to share your experience/perspective

Absence

- Max 3 "joker" hours
- Have a good reason
- Ask us in advance
- Get confirmation
- Make up for it

Conduct

> Be excellent to each other

- Actively participate
- Ask for / give help
- Respect boundaries

Program Week 1

Day	Input	Making
Mon	Intro & DIY-Electronics	Soldering
Tue	Embedded Programming	Tinkering
Wed	Design & Ideation	Prototyping
Thu	Digital Fabrication in 2D	Laser-cutting
Fri	Digital Fabrication in 3D	3D-printing

Program Week 2

Day	Input	Making
Mon	PCB–Layout	Etching
Tue	Documentation & Sharing	Exploring
Wed	DIY–Project Kick–off	Working
Thu	DIY–Project Inputs	Working
Fri	DIY–Project Inputs	Presenting

Program Specials

> At HTU, all participants are welcome

- Week 1, Mon, 07:45, welcome coffee
- Week 1, Fri, 12:00, free open lunch
- Week 2, Fri, 16 - 18:00, grill party

Topics

0. Syllabus & Overview
1. DIY-Electronics & Soldering
2. Embedded Programming
3. Design, Ideation & Prototyping
4. Digital Fabrication in 2D
5. Digital Fabrication in 3D
6. PCB-Layout & Production
7. Documentation & Sharing
8. DIY-Project & Presentation

Structure

Each topic follows this structure

- Motivation, why this is relevant, by example
- Basics, what you have to know, to get started
- Advanced, what brings you further, after basics
- Materials, what we provide to do hands-on tasks
- Tools, which tools you can use to get experience
- Skills, what you can learn by doing it yourself
- Resources, where you find more, quality info

Materials

- Template repo, as a starting point
- Box of tools per team, to be returned
- Making materials, to be used responsibly

* https://github.com/tamberg/fhnw-phycom

Audience

- This course can be attended by students from both, HSI and HTU
- Basic knowledge of either programming or electronics might help
- If you are not sure if this course is for you, consider contacting us

Contact

- Contact us via MS Teams or email thomas.amberg@fhnw.ch